Software Defined Radio based Emulation of SAT-Terrestrial Network Coexistence

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***Abstract*—As cloud applications and 5G use cases increase the demand for additional spectrum, there has emerged interest in spectrum sharing between terrestrial and satellite (SAT) networks. This paper assesses Dynamic Exclusion Zones (DEZ) and Power Control for spectrum coexistence between 5G Networks and SAT Digital Broadcasting Stations (DVB-S2). Relying on a baseband Software Defined Radio (SDR) based emulation on the COSMOS testbed, we propose a spectrum sharing control model that leverages a joint Dynamic Radio Resource Management (RRM) algorithm (Power Control with DEZ) for optimized control. Furthermore, the model combines RRM strategies with artificially generated atmospheric attenuation and path losses for robust control in a standard communication environment. Evaluating our implementation with a ETSI Rural Macro case study, we emphasize our system's effective power control to preserve DVB-S2 Receiver Packet Error Rate (PER) performance, while maintaining a satisfactory 5G network PER. Against varying weather conditions, our results reveal joint RRM strategies can restore link performance within 73% to 99% of the ideal (no-interference) in a single base station (BS) interference scenario and 82% of the ideal against multiple BS conditions.**

***Index Terms*—Spectrum Sharing, Power Control, Dynamic Exclusion Zones, 5G**

I. INTRODUCTION

The development of commercial 5G New Radio (NR) applications has rapidly advanced and challenged the requirement for reliability, low latency, and spectrum utilization in wireless networks [1], [2]. As innovations in bands below 7.125 GHz for Frequency Range 1 (FR1) and in 24 to 52.6 GHz for Frequency Range 2 (FR2) face spectrum insufficiency and mmWave deployment challenges, 7.125 to 24 GHz Frequency Range 3 (FR3) band provides new opportunities for expansion with favorable radio propagation characteristics that enable small cell Base Stations (BS) to expand 2.33 times past its FR2 coverage radius [3]. However, as active and passive satellite (SAT) systems occupy the FR3 band, the challenges of spectrum coexistence and mitigating 5G interference highlight the necessity for spectrum sharing algorithms and Dynamic Radio Resource Management (RRM).

Therefore, it is crucial to evaluate a dynamic spectrum coexistence framework that accommodates multiple networks, prevents obstruction of SAT service performance, and realizes the potential of the FR3 band's characteristics. In this paper, we propose a spectrum sharing control model that facilitates power adjustments via Software Defined Radio (SDR) based emulation, co-located sensor deployment, and intelligent feedback to handle coexistence between baseband SAT and 5G NR transmissions. This model adjusts for atmospheric propagation effects in the FR3 band and 5G's Orthogonal Frequency Division Multiplexing (OFDM) waveforms, where the SAT signal fidelity might be impacted by a mercurial environment.

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Our Contributions: Based on these requirements, this paper makes the following contributions.

1) Emulating a 5G and SAT Digital Broadcasting Station (DVB-S2) interference scenario through SDRs.
2) Creating a joint RRM (Dynamic Exclusion Zone and Power Control) algorithm to maximize SAT reception quality and mitigate 5G interference.
3) Assessing system model performance through a case study emulated on the COSMOS [4] testbed.

The paper is organized as follows: Section II provides Literature Review, Section III presents the Spectrum Coexistence Model, Section IV explains the Emulation Framework, Section V describes the RRM techniques, Section VI details the Rural Macro interference scenario, Section VII presents Results, and Section VIII details Conclusions and Future Work.

II. LITERATURE REVIEW

The opportunities for 5G interference models and efficient spectrum algorithms in the FR3 band have encouraged RRM and 5G spectrum sharing surveys, as documented in studies [5] and [6]. Consequently, this review explores the implementation and challenges of recent RRM simulations and FR3 models.

A. Dynamic Exclusion Zone Simulations

Offering a context-based spectrum sharing model, authors of [7] evaluate RRM policies such as Dynamic Exclusion Zones (DEZ), deploying a computer-simulated emulation of a 5G environment. However, the implementation requires massive server infrastructure and emphasizes 5G cutoffs, hindering FR3 experimentation and spectrum management strategies.

Similarly, DEZ research through Monte-Carlo simulations exemplifies policies in statewide SAT spectrum sharing cases [8], [9]; nevertheless, while the works encourage macroscopic spectrum sharing observation, these emulations generalize the parameters of weather, terrain, and BS allocation. Furthermore, they do not have dynamic BS power control, ignoring coexistence scenarios with partial transmission power. However, our work explores these events through a SDR based emulation, assessing dynamic and physical RRM techniques for scalable commercial off the shelf (COTS) hardware.

B. FR3 Characteristics Modeling

The construction of FR3 scenarios produces intriguing environmental frameworks, where the authors of [10] address FR3 atmospheric and path losses through simulated models. While their research proposes realistic receiver gain losses, emulation constraints such as 50% User Equipment (UE) loading do not consider DVB-S2 waveform's sensitivity in the FR3 environment. Additionally, the authors expand on their environmental framework with receiver antenna patterns and FR3 standardization consistent to 3GPP requirements [11]. However, this approach only accounts for non-line-of-sight (NLOS) path losses, instead of factoring more weather propagation effects or SAT orbital scenarios in FR3. Addressing this, we incorporate a diverse set of atmospheric and path losses during baseband processing, reproducing the SAT downlink for comprehensive SAT coexistence in FR3.

C. Spectrum Sharing and Radio Resource Management

The field of dynamic spectrum access proposes advancements and frameworks involving cognitive radio deployment, along with intelligent power control. Authors of [12] leverage COTS resources and implement a deep reinforcement learning approach. Meanwhile, authors of [13] propose a channel allocation based power control policy and employ RRM to handle BS resources, but both works deploy in a minimalist indoor environment, which limits its utility for FR3 band's outdoor attributes and other RRM solutions like DEZ. The authors of [14] present a licensed shared access testbed for a coexistence scenario between SAT and real/software emulated 5G BS. Comparatively, all works are scalable for SDR based emulation, but they lack validation on the FR3 environment and have loose frequency coordination schemes for rigid 5G NR deployments, which our work seeks to address. Physical setups found in [15] explore spectrum sharing through massive software infrastructures and complex sensor arrays, emphasizing specialized hardware for spectrum sharing. Instead, we utilize SDRs for multi-functional control and observation through Adaptive Energy Detection, showcasing spectrum sharing with reduced hardware and software complexity.

III. Spectrum Coexistence Model

As shown in Fig. 1, the spectrum coexistence model assesses co-located DVB-S2 and 5G downlinks, where RRM control algorithms handle the transmission power and DEZ for the 5G BS. These spectrum sensors provide RRM inputs Signal-to-Noise-Ratio (SNR) and energy (dBm) for a specified channel. Other RRM inputs are the Packet Error Rate (PER) from the 5G UE and DVB-S2 Receiver. To enable coexistence, the spectrum manager takes these RRM inputs for 5G BS

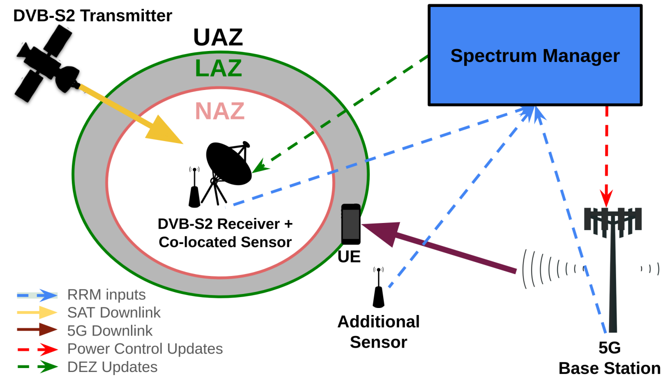


Fig. 1: Spectrum Management Model

transmitter power control updates. The model incorporates one co-located sensor with the DVB-S2 Receiver and another placed between the DVB-S2 Receiver and 5G BS. The co-located sensor provides direct RMM input feedback when the DVB-S2 Receiver's Phased-locked loop is disabled, and the additional sensor supplements an intermediary reading of the 5G BS effects on the DVB-S2 Receiver for more reliable assessment. While Fig. 1 shows a single BS and a single DVB-S2 link, this model can take RRM inputs from a scalable set of spectrum sensors, maximizing informed and data-driven management through 5G and DVB-S2 network readings.

As seen in Fig. 1, the 5G link is located in proximity to the DVB-S2 Receiver's DEZ [16] of No Access Zone (NAZ), Limited Access Zone (LAZ), or Unlimited Access Zone (UAZ). These DEZ deterministically control the 5G network transmission power regulation in a 100% spectrum overlap scenario. Furthermore, the spectrum manager updates the DEZ, affecting transmit power from the 5G BS to the UE through preassigned power control. The spectrum sharing DEZ ensures preservation of the DVB-S2 reception fidelity, while extracting relative performance from the 5G network instead of terminating it through DEZ.

IV. Emulation Framework

Establishing optimized RRM techniques for spectrum coexistence between 5G networks and SAT services in FR3 presents significant infrastructure and resource allocation challenges. As it is infeasible to test with active 5G infrastructure and to develop a fully-realized commercial satellite, we prototype the spectrum coexistence model with open-source emulation of 5G OFDM and DVB-S2 protocols. Although this emulation does not reflect real 5G BS or DVB-S2 transmissions, the SDRs' will transmit over the air on 3.5 GHz, which is downconverted for baseband processing and software accessibility to emulate the FR3 interference characteristics of these networks. Furthermore, to reduce dependency on the COSMOS testbed, our framework allows geographical alterations of the DEZ, replicating and reproducing artificial variables and physical distances without relying on COSMOS's physical attributes. Thus, baseband processing and SDR attenuation will account for these deficiencies, replicating artificial geographic setups in [17], [18]. Furthermore, in

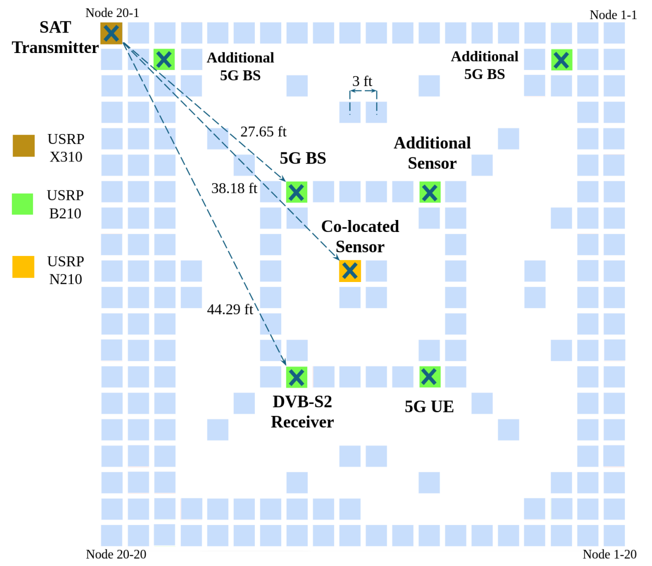


Fig. 2: Implementation Scenario on COSMOS Grid

the results, we only consider the effect of the 5G BS downlink, as the BS transmitter's interference does not require mobile nodes. Without loss of generality, the framework presented in the rest of the paper includes capability for uplinks.

A. Hardware and Software Setup

Experiments are performed in COSMOS testbed, utilizing COTS SDRs to incorporate RRM strategies. COSMOS grid deploys terrestrial 5G OFDM BS, UE, and DVB-S2 reception through the B210 USRPs. For single interference, a single B210 27.65 ft. away from the SAT transmitter is utilized, while multiple interference scenarios included additional B210s as seen in Fig. 2. Modeling the SAT transmission and its distant orbital transmission, an X310 SDR node is utilized. Meanwhile, a N210 SDR node is deployed for the co-located DVB-S2 sensor, along with a B210 SDR node for sensor and spectrum management between the 5G and DVB-S2 network. The physical distance between nodes is 3 ft.; however, the network software and artificial characteristics are supplemented through GNU Radio 3.10 on Linux Ubuntu 22.04 OSs. Thereby, the framework is artificially reproducible regardless of radio configuration, operating system, or larger quantities of radios.

B. 5G Waveform

Our model uses a GNU Radio based 64-bin OFDM, representing a stable and flexible structure against frequency-selective fading. In addition, the OFDM reinforces 3GPP standard 5G NR experimentation for stronger validation of narrowband interference in a dynamic spectrum context, permitting scalable scenarios and waveform modifications. Due to processing and open-source constraints, the emulation of 5G BS network is modulated by 5G NR standards of Quadrature Phase-Shift Keying for header information and 16 Quadrature Amplitude Modulation for the payload information. Extending to larger deployments, the bandwidth size and symbol rate are limited to the sampling rate associated with the SDRs' computation. However, a sync word is utilized to follow 5G NR network standards in synchronization mismatch.

C. Digital Video Broadcasting Waveform

We incorporate a DVB-S2 transmission through GNU Radio, which allows variability in modulation and forward error correction. For handling potential interference, the employment of a physical layer frequency correction resembles the DVB-S2 Physical Locked-layer that handles time synchronization, forward error correction, and frame processing for SAT performance. Modulations for the DVB-S2 are conducted in 8 Phase-Shift Keying with $\frac{3}{5}$ low-density parity error correction on normal frames to match ETSI standards. For a video source, a looped MPEG Transport Stream is transmitted, handling a practical DVB-S2 transmission.

D. FR3 Modeling

The FR3 weather modeling algorithm supplements baseband radio waves with artificial noise through path losses and FR3's atmospheric losses: cloud, snow, rain, oxygen, and water vapor. To approximate these losses, the usage of 3GPP [19] and ITU-R tables is implemented for FR3's atmospheric loss. The path loss equation (in dB) found in [10] and [19] is a LOS model assuming no obstacles, distance of d (meters) between transmitter and receiver, frequency f_c of the given signal, the environment's zero-mean shadow fading loss $\mathcal{X}$, and standard deviation σ_{LOS} to produce

$$PL(d) = PL_{LOS}(f_c, d) + \mathcal{X}(0, \sigma_{LOS}). \tag{1}$$

This equation mirrors the open-space of the COSMOS testbed, facilitates an open outdoors environment, and provides sufficient parameters found in other works like [19]'s Table 7.4.1 to accurately calculate loss in both the 5G and SAT networks.

Accounting for the DVB-S2 downlink path loss to the receiver and 5G signal attenuation to UE, a gain and environmental model evaluates temporal effects of the SAT downlink through

$$A_{Fall}(\theta, h, T, R, f_c) = (\mathbf{1}_a A_{env}(f_c, R))(A_{atm}(h, f_c))G(\theta),$$

where h, T, R, and f_c denote the attenuation parameters of height (kilometers) in effective slant range, temperature (Kelvin), rain rate (mm/h), and baseband processing frequency. For $A_{env}(f_c, R)$, the attenuation value was produced by the frequency's ITU-R table, and $\alpha \in \{0, 1, 2, 3\}$ is a selection for ideal, day, rain, or snow scenario, which multiplies with the general atmospheric attenuation $A_{atm}(h, f_c)$. For the ideal case, attenuation value excludes factors associated with clouds and humidity. $G(\theta)$ (in dBi) is provided by

$$G(\theta) = \begin{cases} 40 - 25\log(\theta), & 6^\circ \leq \theta \leq 48^\circ \\ -2, & 48^\circ \leq \theta \leq 180^\circ \end{cases} \tag{2}$$

to produce an emulated gain associated with boresight angle of the SAT receiver and transmitter pair [20]. Through a designated SAT downlink, a slant range and $10^\circ < \theta < 90^\circ$ elevation angle calculation is produced. If there are more complex environments, the calculations can be integrated into a tuned DVB-S2 non-geostationary model for spectrum interference.

V. Radio Resource Management

A. Dynamic Exclusion Zones

The DEZ algorithm performs 5G BS power adjustments through an artificial boundary. During the DVB-S2's downlink, this algorithm updates every minute, using only RRM inputs at the DVB-S2 Receiver to emulate primary user interference management. Its updates are only validated every minute to account for 30 second resets of DVB-S2 Phase-Locked loop and SNR data collection during significant interference.

Upon registering an inadequate SNR value or physical layer frequency error, the receiver updates the LAZ and NAZ to deter 5G BS interference. Through a procedural sequence, the spectrum manager iterates through radius (Δr) kilometer updates, continuing until the 5G BS transmission is terminated or the DVB-S2 signal is distinguishable. The calculated shifts from the LAZ and NAZ radii are updated for preassigned power outputs. Thus, the dynamic feature only regulates the zones, whereas preassigned transmit outputs emulate previous DEZ implementations.

For the radii assignments, the setup consists of

$$r_{BS} \geq \begin{cases} \Delta r_{NAZ} + r_{NAZ}(n) = r_{NAZ}(n+1) \\ \Delta r_{LAZ} + r_{LAZ}(n) = r_{LAZ}(n+1), \end{cases} \tag{3}$$

where Δr_{LAZ} and Δr_{NAZ} are the initial radii updates assigned, r_{LAZ} and r_{NAZ} refer to the radii of the respective zone based on n step increments, and the r_{BS} vector refers to the two element BS threshold that changes transmission output once eclipsed. The following radii will converge to the 5G BS location and prohibit transmission or guarantee a safe DVB-S2 Receiver SNR with limited 5G transmission.

B. Power Control

The co-located sensor utilizes Energy Detection across the 5G BS bandwidth, separating readings into 10 MHz channels during a 100% overlap scenario. Since the sensor accounts for the DVB-S2 Receiver and 5G BS, Signal-to-Interference-Noise-Ratio (SINR) readings become distorted by additional 5G signal power. As an adjustment, an Exponential Moving Average (EMA) formula accounts for E_{avg} or averaging Y_t energy (in dB) of N channels in i iterations, multiplying by weight μ to prefer either future and current value at t time:

$$E_{avg} = \frac{1}{N}(\mu \sum_{i=1}^{N} |Y_t[n]|^2 + (1-\mu) \sum_{i=1}^{N} |Y_{t-1}[n]|^2). \tag{4}$$

Equation 4 describes the EMA observation input used in a Linear Quadratic Regulator (LQR) based power control algorithm to handle interference [21]. Specifically, the algorithm restricts E_{avg} from falling below a threshold set to a relative gain of -70 dB or COSMOS's noise floor. While it cannot detect a specific network, it provides a limited scope of the 5G transmission, reducing spectrum sensors' distortion.

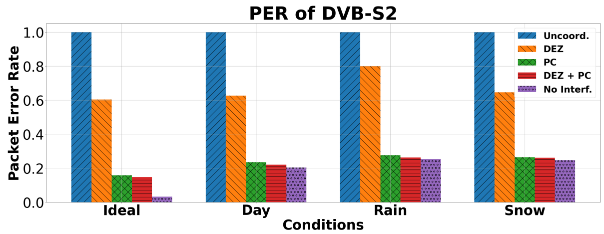
C. Joint Power Control with Dynamic Exclusion Zone

The joint RRM strategy combines LQR based power control with DEZ through millisecond calculations, as shown in Algorithm 1. Specifically, the spectrum sharing model becomes a discrete-nonlinear system controlled by BS transmit power θ_p from $[0, 1.0]$. The observation or state vector $x[n]$ relies upon the millisecond RRM inputs (see Fig. 1). This permits consistent measurements during DVB-S2 Receiver failure but ensures relative 5G downlink performance. Meanwhile, the control input $u[n]$ for θ_p is calculated for LQR's K gain matrix. However, while calculations are fast, power control adjustments are still a second, due to the SDR computational resources and maintaining reliable control sequences. The control input is an unlimited transmission output P_T or limited transmission power output P_{LAZ}. The LQR based power control incorporates weighted A (input) and B (input-to-state) matrices, highlighting transmission power's influence on $x[n]$. To avoid 5G downlink termination, the LQR places a higher penalty on state cost penalty matrix Q's BS PER.

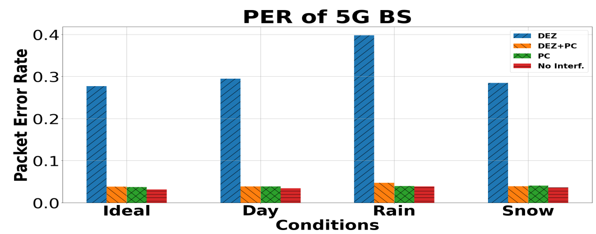
If an observational iteration of the DVB-S2 Receiver fails, co-located sensor feedback indicates a lock on the mechanism, accounted through the LQR's adjustment of Q matrix. If the DVB-S2 Receiver locks in its given iteration, then larger penalties are placed on Q's sensor energy readings and PER. This emphasizes immediate sensitivity to locks, promoting swifter actions to mitigate real-time loss of information and limited feedback. The LQR policy in Algorithm 1 employs a Discrete Algebraic Ricatti Equation (DARE), which is processed by the spectrum manager; thereby, the LQR policy's successive updates are handled by the spectrum manager to converge to a controllable spectrum management scenario.

Algorithm 1 LQR Control for Joint Power Control

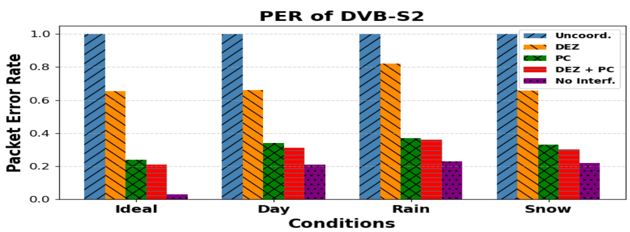
$u[n] \leftarrow \max(0, \min(1.0, u[n]))$ # Initialize control vector
$\theta_p \leftarrow 1.0$; $u[0] \leftarrow \theta_p$ # Initial control input
$x[n] \leftarrow zeros()$ # Initial state vector
A, $B \leftarrow inputs()$ # Input and input-to-state matrices
Q_{normal}, $Q_{\text{lock}} \leftarrow states()$ # State cost penalty matrices
$R \leftarrow diag(1)$ # Set single input-cost matrix
while system is active **do**
  Measure state: $x[n] \leftarrow \{Energy, SNR, PER, lock\}$
  if $lock$ condition is true **then**
    $Q \leftarrow Q_{\text{lock}}$
  end if
  Solve P by solving $DARE(A, B, Q, R)$
  Compute LQR gain matrix: $K \leftarrow R^{-1} B^T P$
  Compute new control input: $u[n] \leftarrow -Kx[n]$
  Incorporate LAZ updates via Equation 3
  if $r_{LAZ}(n) \geq r_{BS}$ **then**
    Set $u[n] \leftarrow u[n] * P_{LAZ}$ # LAZ power threshold
  else
    Set $u[n] \leftarrow u[n] * P_T$ # UAZ power threshold
  end if
  Update system state on $u[n]$ and get feedback
end while



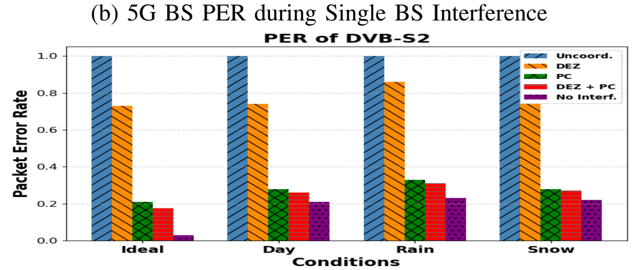
(a) DVB-S2 Receiver PER with Single BS Interference (10 km)



(b) 5G BS PER during Single BS Interference



(c) DVB-S2 Receiver PER with 3 BS (10 km, Same Distance)



(d) DVB-S2 Receiver PER with 3 BS (10, 12, and 15 km)

Fig. 3: PER of DVB-S2 Receiver for Single BS and 3 BS with Different BS Placement

VI. Case Study: Rural Macro Region

The losses characterized by the FR3 environment derive from [19]'s Rural Macro parameters at 12.7 GHz. Our emulation of the DVB-S2 SAT downlink baseband signal is handled at the DVB-S2 Receiver, during a 100% overlap FR3 interference scenario. Its 10 minute downlink pass relays a MPEG transport stream in non-geostationary orbit. The baseband processing will account for the distance between the SAT transmitter and receiver, attenuating the downlink signal.

Applying the radio mapping features of COSMOS Grid [17], [18], the ETSI Rural Macro case study experimentation demonstrates how a scalable 5G BS and DVB-S2 downlink placement impacts FR3 spectrum coexistence. To compare insights with previous FR3 simulations, the BS and DVB-S2 Receiver parameters are emulated for the Rural Macro with Rician Fading model provided by the authors of [7] and demonstrated in Fig. 4; thereby, parameters such as humidity, temperature, rain rate, and water vapor density derive from the Rural Macro LOS path loss model. All BS are located 1 km from the UE, following ETSI model. The only interference considers the DVB-S2 link from the 5G BS transmission, so BS will be placed 10, 12, and 15 km from the DVB-S2 Receiver.

The emulation utilizes the spectrum sensor located to sense the channel overlap, facilitating a centralized computer (spectrum manager) to each of the links' received signals. For the LAZ and NAZ, we initialize the radii to 1 km and 0.5 km in relation to previous DEZ papers' boundaries; the preassigned $\theta_p \in \{0, 0.7, 1\}$ further delves and compares [16]'s dBm outputs. The sequential updates to the radii are based on the SNR values of the DVB-S2 co-located sensor and follow iterative increments of 1 km and 0.5 km respectively. If the LAZ is approached, the 5G BS transmission gain will be limited to $P_{LAZ} = 0.7$ to ensure the SNR of the DVB-S2 Receiver is an adequate signal around 9.0 dB, maintaining a qualitatively consistent MPEG Transport Stream. Through post-processing, the RRM strategies will be parallelized by SDR operations, performing real-time analysis and transmitter power control over 3.5 GHz transmission. To assess performance, PER for the DVB-S2 Receiver and the 5G UE are measured.

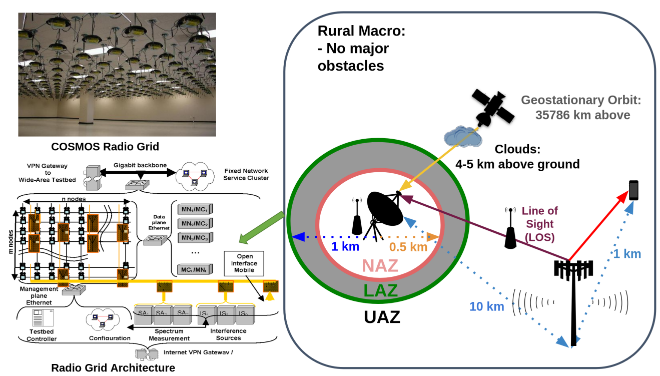


Fig. 4: Rural Macro Setup Example on the COSMOS Grid

VII. Results

Fig. 3 shows the PER for the DVB-S2 and 5G BS links respectively for different weather conditions (Ideal, Day, Rain, Snow), the relative performance of each of the RRM techniques (DEZ, power control [PC], joint power control [DEZ + PC]), the uncoordinated spectrum sharing (Uncoord.), and no interference (No Interf.) scenarios. From Fig. 3(a), the attenuation by non-ideal weather results in a significant degradation of DVB-S2 20% PER performance, highlighting FR3 band's impact during single BS interference. This is due to attenuation from day, snow, or rain preventing SAT downlink accessibility below an elevation angle 20°. The RRM technique of DEZ marginally improves DVB-S2 PER performance, while the power control has significant effect as shown in Fig. 3(a), 3(c), and 3(d). Regardless of weather, the implementation of power control over DEZ provides a nearly 40% PER improvement in single BS interference and 30% PER improvement in multiple BS interference. Therefore, power control performs better against DEZ spectrum management, since it updates several times a second as opposed to DEZ that updates every minute; it provides more immediate control, without depending on longer durations of the DVB-S2 Receiver's Phase-locked loop sensor. Furthermore, the DEZ's preassigned power outputs are not situated for a nuanced Rural Macro setup, preventing gradually 5G BS adjustment; comparatively, joint power control almost restores PER performance close to the No Interf. scenario. Thereby, FR3 spectrum sharing depreciates the DVB-S2 signal, highlighting preservation of requisite signal fidelity requires better sensor deployment and quicker updates found in joint RRM. Therefore, more optimally placed sensors could enhance signal fidelity with reliable measurements. In denser urban networks, this requires strigent policy and higher computation for sensor storage, although the LQR algorithm and updates are relatively light; the sensors' streamlining data will require stronger authenticating and processing to ensure a more equitable calculation with increasing UEs and BSs. Figure 3(b) shows the impact of such RRM for spectrum coexistence on the 5G BS link. It is observed that once again power control greatly improves the PER performance and relative 5G BS link performance is obtainable for all weather conditions. Overall, by observation, joint RRM handles scalable complexity for both 5G BS and DVB-S2 PER performance with respect to No Interf. scenario.

VIII. Conclusion and Future Work

This paper demonstrated the capability of joint RRM techniques including LQR based power control and DEZ by SDR based emulation. Through experimentation on COSMOS testbed, joint RRM fundamentally preserved DVB-S2 PER, while ensuring sufficient 5G BS PER. Relying on the sensors, the SDRs' observational data delineates how spectrum sharing policy not only necessitates more reliable data collection outside of SAT systems, but also demands faster computations to handle complex interference scenarios.

Future insights into FR3 require the modality of SDR based emulation. To improve emulation and advance spectrum sharing policy, our next experiments will utilize a Transceiver Design to upconvert SDRs' transmission to 12 GHz. Thereby, the SDR in higher frequencies and path will show how spectrum sharing policy demands computational optimal algorithms for transmission power control. Therefore, this transceiver design can provide a versatile set of applications and modeling, which can be incorporated into the future considerations of the SDR based Emulation between SAT and 5G BS.

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